

Lavinium WCET Analyzer and Optimizer

POLIMI | ISOLDE | T4.3 / D4.5 Section 3.3.2 | Version ISOLDE-D4.5-2026.07

Schema Version			0.9
Basic Information			
IP Name		Lavinium WCET Analyzer and Optimizer	
Version		ISOLDE-D4.5-2026.07	
Status		Demonstrated	
Comment		Draft metadata release derived from ISOLDE D4.5. The value in basicInfo.version identifies this card release, not necessarily the partner’s product or repository tag. Partner confirmation is required before publication.	
Provider	Item 1 / Organization Name	Politecnico di Milano	
Provider	Item 1 / Organization URL	https://www.polimi.it/	
Provider	Item 1 / Organization Alias	• POLIMI	
License		Apache-2.0	
Repository URL		https://github.com/HEAPLab/Lavinium	
Project		ISOLDE	
Work Item		T4.3 / D4.5 Section 3.3.2	
IP Type	Category	• Software Tool • Compiler • Timing Analysis Software • Domain Specific Optimization and Code Generation Tools	
IP Type	Domain	• Safety	
IP Type	Comment	LLVM-based function-level WCET optimization	
IP Type	Service	no	
System-Level Information			
Architecture		RISC-V	
Processor		RISC-V targets supported by LLVM 17 and LLVMTA	
ISAExtension		Target-dependent	
Board		Validated on a RISC-V MCU	
SoC		Real-time RISC-V software target	
SW Dependencies		LLVM 17, LLVMTA and their RISC-V backend/toolchain dependencies	
Requires Specialized Tool		Yes. LLVM 17 and the LLVMTA WCET analysis stack.	
Interfaces and Artifacts			
CLI		• Compile LLVM IR through the Lavinium optimization flow	
API		• LLVM compilation-pipeline integration • Function-level optimization rescheduler	

Input Artifacts	Item 1 / Artifact	Program representation
Input Artifacts	Item 1 / Format	LLVM IR
Input Artifacts	Item 2 / Artifact	Optimization-pass set and exploration policy
Input Artifacts	Item 2 / Format	Configuration
Input Artifacts	Item 3 / Artifact	WCET analysis target configuration
Input Artifacts	Item 3 / Format	LLVM-TA configuration
Output Artifacts	Item 1 / Artifact	Optimized program representation
Output Artifacts	Item 1 / Format	LLVM IR
Output Artifacts	Item 2 / Artifact	RISC-V executable
Output Artifacts	Item 2 / Format	Binary/ELF
Output Artifacts	Item 3 / Artifact	WCET estimates and exploration results
Output Artifacts	Item 3 / Format	Analysis data
Deployment		
Bundle	Open-source Git repository with installation and usage guide	
Project-Specific Features		
Purpose	Explore optimization-pass ordering to reduce per-function WCET.	
Validated Capabilities	• LLVM 17 LLVM-TA integration • Deterministic and stochastic exploration • Local-minimum search • Exploration-time/performance trade-off • RISC-V binary generation and execution	
Limitations	• Demonstrator integration study was still ongoing at the D4.5 cut-off.	
Public Repository Status	Public	
Source Basis	Deliverable	ISOLDE D4.5 - Software for Hardware Design and Validation Implementation
Source Basis	Section	3.3.2 Optimizing WCET with LAVINIUM Toolchain: A Scheduling Solution for Improved Reliability
Source Basis	SW IP Card Model	Schema v0.9 with basicInfo, systemLevelInfo, interfaces, deployment and features
Requirements Summary	Count	14

Requirements Summary	Ids	• POLIMI-WCET-01 • POLIMI-WCET-02 • POLIMI-WCET-03 • POLIMI-WCET-04 • POLIMI-WCET-05 • POLIMI-WCET-06 • POLIMI-WCET-07 • POLIMI-WCET-01 • POLIMI-WCET-02 • POLIMI-WCET-03 • POLIMI-WCET-04 • POLIMI-WCET-05 • POLIMI-WCET-06 • POLIMI-WCET-07
Metadata Review	Partner Review Required	true
Metadata Review	Items	• Replace the metadata release identifier with the exact product/repository version where available. • Confirm public repository or product URL. • Confirm licensing and redistribution wording. • Add a tool-specific TRL only if the partner has approved it.